Scanned Code Report

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Code Info

Developer Scan

Scan ID 23	Date September 12, 2026
Organization RigoBlock	Repository v3-contracts
Branch fix/hype-same-block-lock	Commit Hash db2816f0...fb66dfe9

Contracts in scope

contracts/protocol/libraries/HyperliquidLib.sol

Code Statistics

Findings 3	Contracts Scanned 1	Lines of Code 111

Findings Summary

Total Findings

- High Risk (0)
- Medium Risk (3)
- Low Risk (0)
- Info (0)
- Best Practices (0)

Code Summary

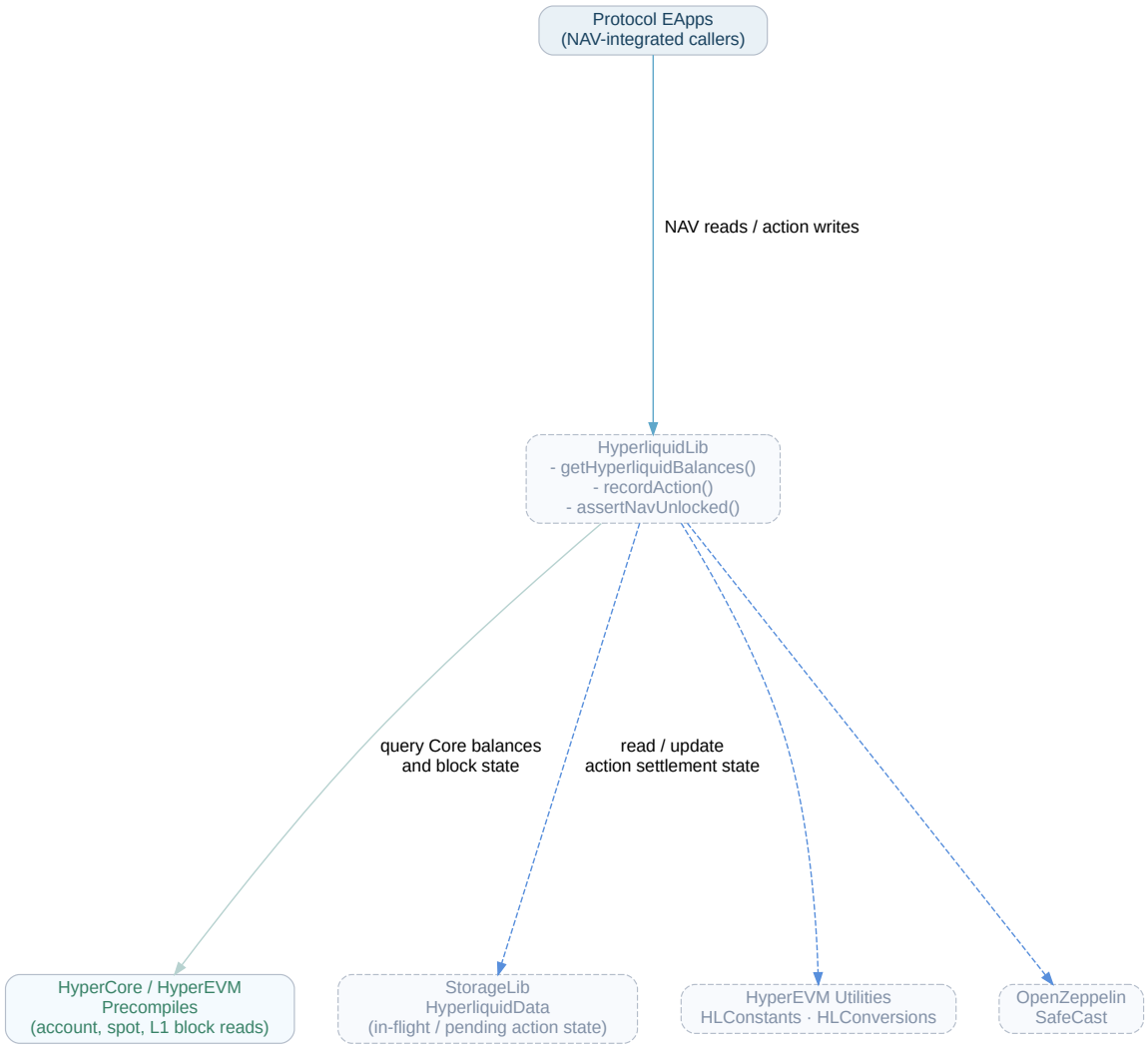
HyperliquidLib is a specialized integration library designed for the Rigoblock asset management protocol to interface with Hyperliquid on HyperEVM. It facilitates accurate Net Asset Value (NAV) accounting and state synchronization between the EVM execution environment and the underlying HyperCore layer. The library reads margin summaries from perpetual markets and token balances from spot markets via precompiled contracts, consolidating them into a unified USDC-denominated valuation for a given pool account.

To address asynchronous transitions between EVM blocks and HyperCore settlement, HyperliquidLib tracks in-flight transactions and pending spot sends using a composite block structure that pairs L1 and EVM block heights. It includes balance adjustments within the active block to prevent miscalculations and enforces a settlement time window to prevent stale NAV computations while cross-layer actions are settling.

Main Entry Points

Because HyperliquidLib is an internal library providing core math, precompile integration, and accounting helpers for other protocol contracts, it does not expose any direct public or external state-modifying functions to external actors.

Code Diagram



1 of 3 Findings

contracts/protocol/libraries/HyperliquidLib.sol

Unrecorded marketable orders can leave NAV writable against stale HyperCore account value

Medium Risk

`HyperliquidLib` makes settlement locking entirely dependent on the pool having previously called `recordAction`. The lock state is represented only by `HyperliquidData.lastActionCompositeBlock` and `lastActionTimestamp`, both of which are updated exclusively in `recordAction`.

```
function recordAction(int256 amount, bool isSpotSend) internal returns (uint64
pendingBefore) {
    HyperliquidData storage data = StorageLib.hyperliquidData();
    uint256 compositeBlock = _composeBlockNumber();
    // ...
    data.lastActionTimestamp = uint48(block.timestamp);
    // ...
}

function assertNavUnlocked() internal view {
    HyperliquidData memory data = StorageLib.hyperliquidData();
    if (data.lastActionCompositeBlock == 0) return;
    if (data.lastActionCompositeBlock == _composeBlockNumber()) return;
    require(block.timestamp >= data.lastActionTimestamp + _SETTLEMENT_WINDOW, NavLocked());
}
```

The supplied integration context states that marketable limit orders and cancels do not record a Hyperliquid cross-layer action. Consequently, a marketable order that is accepted by CoreWriter can change the pool's HyperCore account value, including through execution fees, realized PnL, or an immediate fill, without updating `lastActionCompositeBlock` or `lastActionTimestamp`.

Until the HyperEVM margin-summary precompile exposes the resulting Core state, `getHyperliquidBalances` can return the pre-trade `accountValue`. Unlike recorded EVM-to-Core deposits, there is no same-block adjustment for the order-induced value change, and unlike recorded deposits or spot sends, there is no post-block settlement lock. A NAV-writing pool action can therefore use an overstated or understated USDC NAV. Depending on the trade outcome, a share holder can redeem at an overstated NAV after a loss or a depositor can mint shares at an understated NAV after a gain, transferring value to or from other pool shareholders when the delayed Core state becomes visible.

```
int256 perpValue = int256(
    PrecompileLib.accountMarginSummary(HLConstants.DEFAULT_PERP_DEX, account).accountValue
);

int256 totalUsdcValue = perpValue + spotValue;

bool recentAction = _hasRecentAction();
if (recentAction) {
    totalUsdcValue += StorageLib.hyperliquidData().inFlightAmount;
}
```

This calculation only compensates a prior non-spot action that was explicitly recorded. A marketable order that bypasses `recordAction` has neither an in-flight adjustment nor a settlement delay, despite being capable of changing `perpValue` asynchronously.

Severity Note:
- That marketable limit orders and cancels skip recordAction is taken from the supplied integration context, not from a call site in this file.
- Material impact assumes the precompile lag (typically same EVM block after CoreWriter) overlaps a NAV write and that the order's PnL/fees are large enough to matter versus pool NAV.

2 of 3 Findings

contracts/protocol/libraries/HyperliquidLib.sol

A later Core action re-enables same-block NAV reads while an earlier action remains unsettled

Medium Risk

`recordAction` retains an in-flight adjustment only for the most recently recorded composite block. When a new EVM/composite block records another Core action, it unconditionally discards the prior block's `inFlightAmount` and replaces `lastActionCompositeBlock` with the new block:

```
if (data.lastActionCompositeBlock != compositeBlock) {
    data.inFlightAmount = 0;
    data.pendingSpotSend = 0;
    data.lastActionCompositeBlock = compositeBlock;
}
```

The settlement guard then immediately permits NAV-sensitive operations in the new block, solely because the most recent action was recorded in that same block:

```
if (data.lastActionCompositeBlock == _composeBlockNumber()) return;
```

This creates an unsafe sequence if an earlier non-spot action has not yet appeared in the Core reader precompiles. For example, assume action A deposits 60 USDC from HyperEVM to Core in composite block N. The pool's EVM USDC balance is reduced immediately, while `inFlightAmount` temporarily restores the missing Core value for same-block NAV reads. If the Core reader is still stale in block N+1, an authorized operator can record a second Core action B in N+1. Recording B clears the 60-USDC adjustment for A, stores only B's adjustment, and causes `assertNavUnlocked()` to return through its same-block exception.

`getHyperliquidBalances` then adds only the adjustment for B:

```
bool recentAction = _hasRecentAction();
if (recentAction) {
    totalUsdcValue += StorageLib.hyperliquidData().inFlightAmount;
}
```

The resulting NAV omits A even though the corresponding USDC has already left the pool's EVM balance. A mint, burn, share-price update, or other NAV-sensitive action executed in block N+1 can therefore use materially understated or otherwise stale pool value. The intended post-block settlement lock is bypassed before its time-based condition is evaluated.

Severity Note:
- Material harm assumes HyperCore precompiles can lag beyond one EVM block, which the integration documents as an edge case rather than proving it cannot happen.
- A second authorized Core action in a later composite block is required; without it assertNavUnlocked still enforces the 16-second window.

3 of 3 Findings

contracts/protocol/libraries/HyperliquidLib.sol

The fixed 16-second settlement lock can expire while Core reader balances are still stale

Medium Risk

The post-block settlement guard is based exclusively on elapsed EVM timestamp time. Once the composite block changes, the same-block correction is no longer included by `getHyperliquidBalances`; after 16 seconds, the guard permits NAV-sensitive execution without checking whether the relevant Core precompile values now include the recorded action:

```
function assertNavUnlocked() internal view {
    HyperliquidData memory data = StorageLib.hyperliquidData();
    if (data.lastActionCompositeBlock == 0) return;
    if (data.lastActionCompositeBlock == _composeBlockNumber()) return;
    require(block.timestamp >= data.lastActionTimestamp + _SETTLEMENT_WINDOW, NavLocked());
}
```

The in-flight correction is separately limited to exact composite-block equality:

```
function _hasRecentAction() private view returns (bool) {
    uint256 lastComposite = StorageLib.hyperliquidData().lastActionCompositeBlock;
    return lastComposite != 0 && lastComposite == _composeBlockNumber();
}
```

Consequently, an EVM-to-Core deposit can be absent from both the pool's EVM USDC balance and the raw Core reader response after the composite block changes. During the first 16 seconds this is blocked, but once `block.timestamp >= lastActionTimestamp + 16 seconds`, the library accepts the stale raw reader state regardless of whether HyperCore has actually processed the action.

The supplied integration context explicitly identifies delayed CoreWriter processing, sequencing delays, and HyperEVM big-block operation as conditions under which a Core reader response can remain stale for longer than the selected settlement interval. Under those conditions, a subsequent mint can receive shares at an understated NAV after a pending EVM-to-Core deposit, while the opposite accounting direction can cause redemptions or other NAV validations to use an overstated value. The discrepancy can be as large as the unsettled transfer amount rather than merely a rounding difference.

Severity Note:
- Treating Core reader staleness beyond lastActionTimestamp + 16s as reachable depends on integration/sequencing context, not on a proof inside this library that settlement can exceed the window.

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